

Representation as a Computational Resource in Quantum Compilation

Approximate recoverability tradeoffs and model-specific fault-tolerant consequences

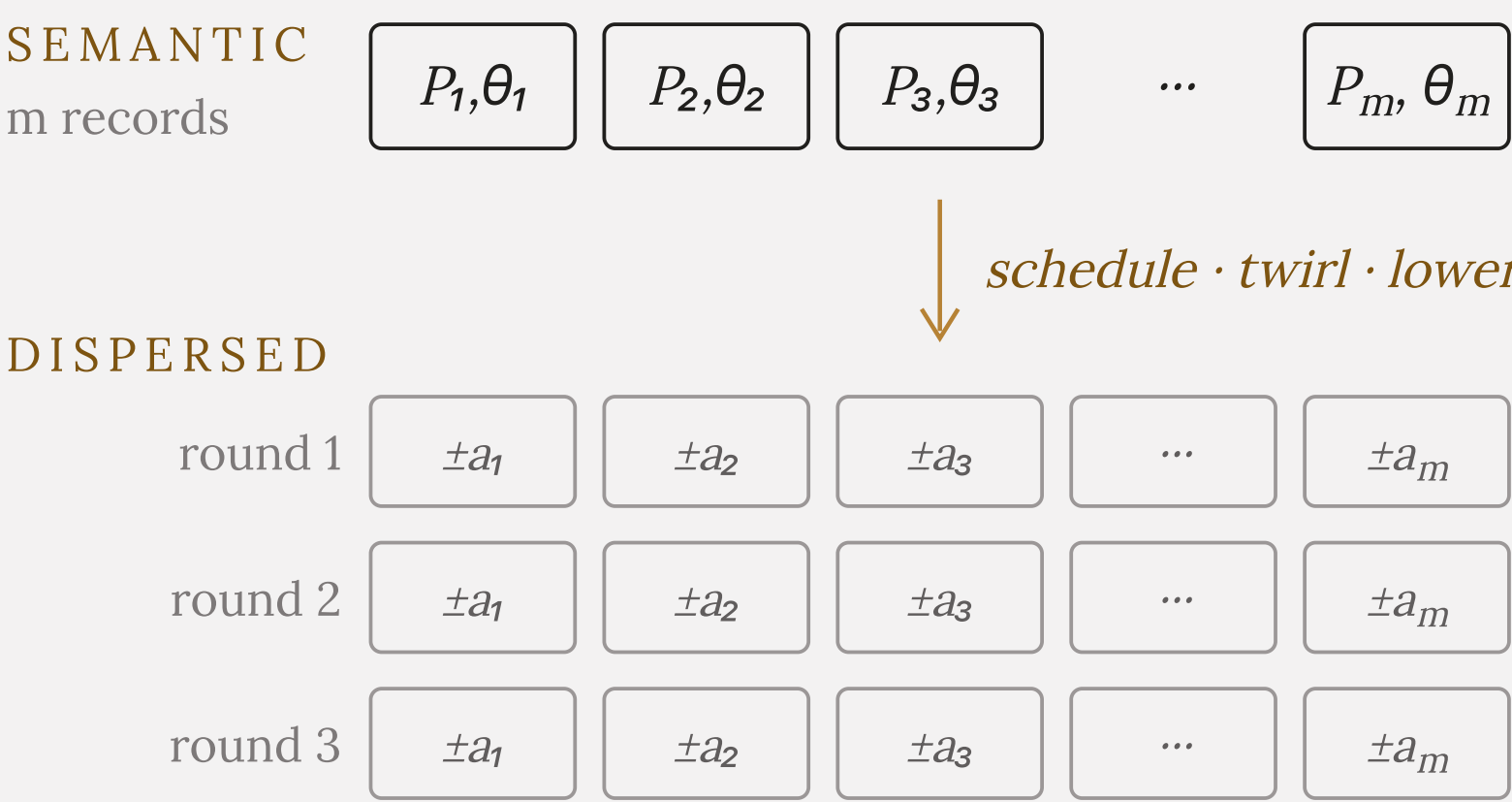
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01 Compilation destroys structure

Fault-tolerant pipelines deliberately scatter a quantity that was once written in one place — and the scattering is correct, standard, and desirable in each case:

- **Scheduling.** Variational and time-dependent layers split one accumulated phase across many rounds.
- **Synthesis.** An angle becomes a Clifford+ T word in which no single gate carries the angle.
- **Randomized compiling.** Each rotation is twirled into sign-randomized fragments.

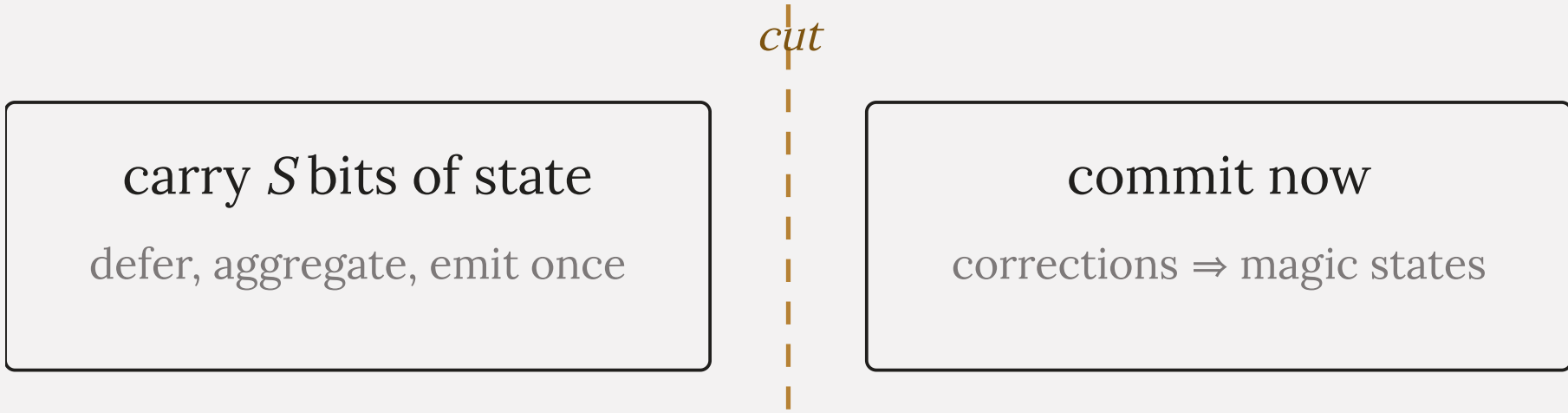
We ask what it costs a *downstream* compiler — a cloud transpiler, a control-stack stream optimizer — to put the quantity back together, and show the price is payable in exactly two currencies: **bits of compiler state**, or **magic states**.



no record — and no prefix — determines the edge total

$$x_j = \sum_t s_j^{(t)} a_j^{(t)} \bmod Q$$

AT ANY CUT



One commuting phase layer, lowered into r contiguous rounds of grid residues. At every cut the scattered information is either carried across in state or committed to output and repaid downstream.

02 A budget-explicit model

Every channel that would let a restarted compiler recover the aggregate is charged in bits — no gate count is ever converted into a bit value:

control	store
live window	parameters
intermediate representation	

Recovery is certified, not asserted. The representation-aware pipeline uses bounded recognition, aggregation certificates and conservative fallback; a measured cell becomes numerical only once a symbolic or dense certificate returns `completed_valid`. Unsupported is never conflated with incorrect, and bridge loss is reported apart from compiler quality.

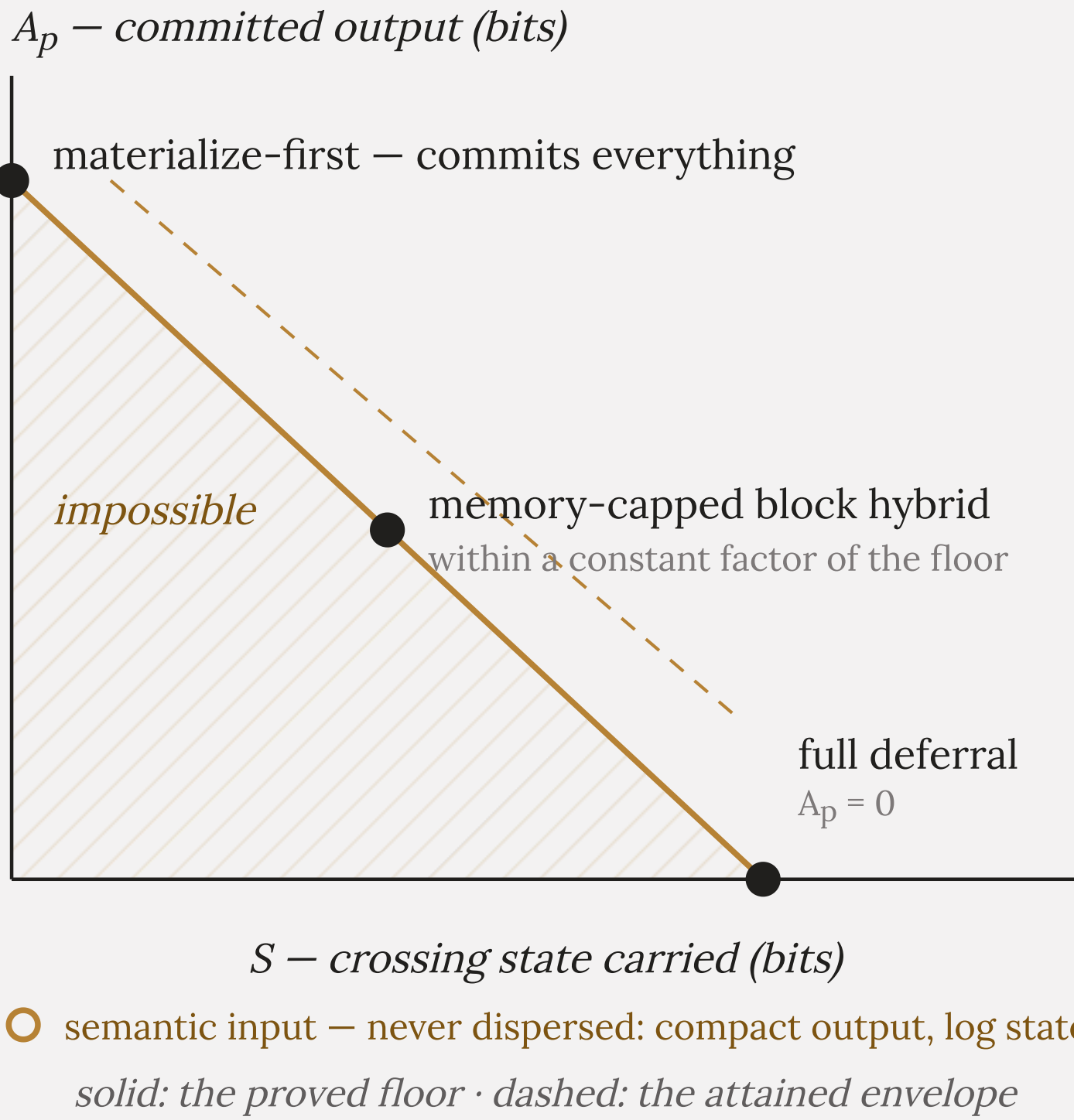
03 The frontier

THEOREM

$$A_p + (2p - 1) S \geq (1 - \delta) m \log_2 K_\epsilon - h_2(\delta)$$

for any compiler making p forward passes and failing with probability at most δ

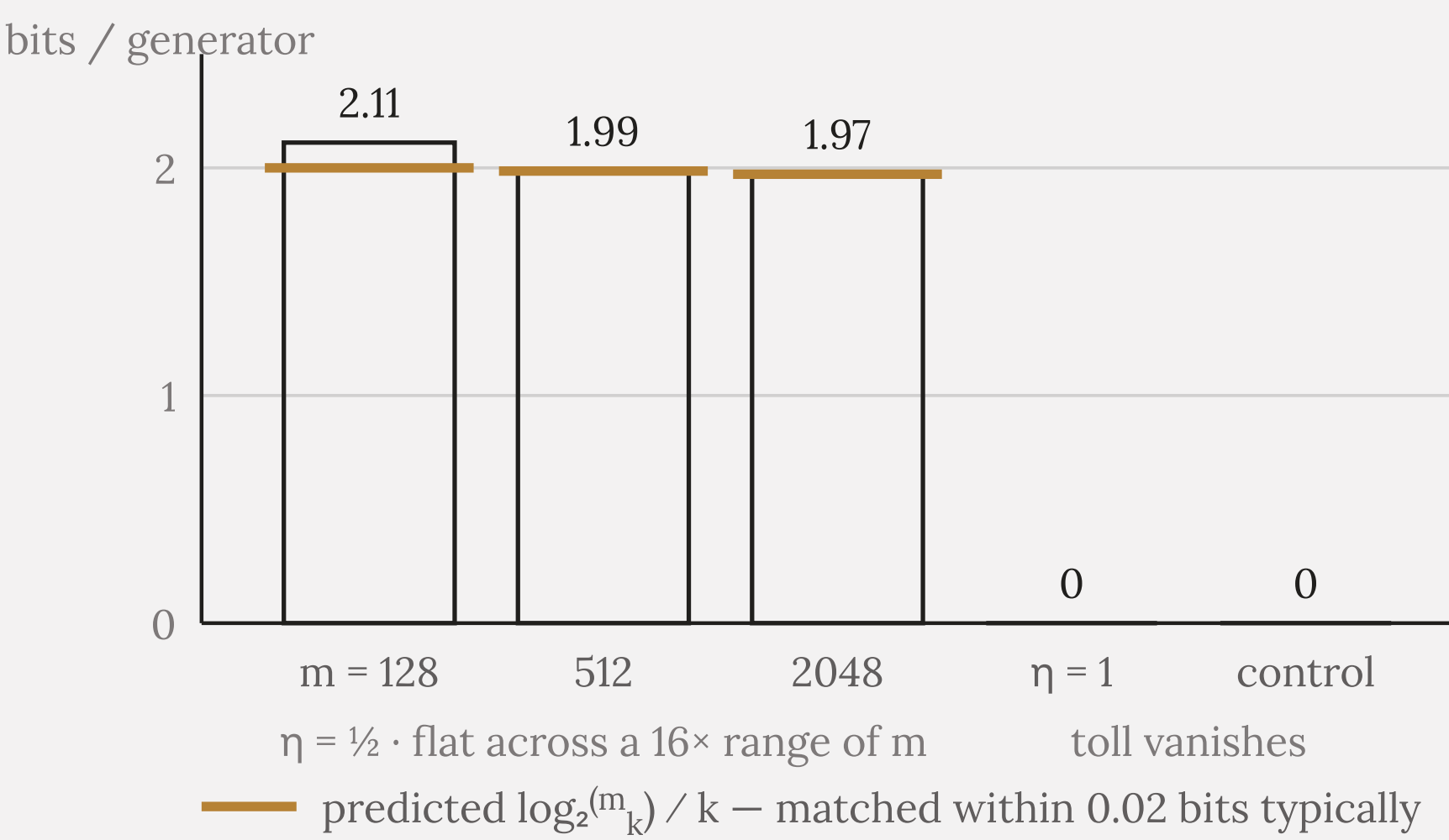
A_p is committed output, S the serialized crossing state, and $K_\epsilon = \Theta(1/\epsilon)$ the number of distinguishable settings on the Ross–Selinger synthesis grid. The two cannot both be small. The bound is carried by the schedule alone, so the folded compile-time presentation of randomized compiling inherits it verbatim — nothing depends on tracking Pauli frames.



Everything below the solid line is unreachable for a compiler fed the dispersed stream; a block compiler reaches the line up to a constant factor.

04 Not at par

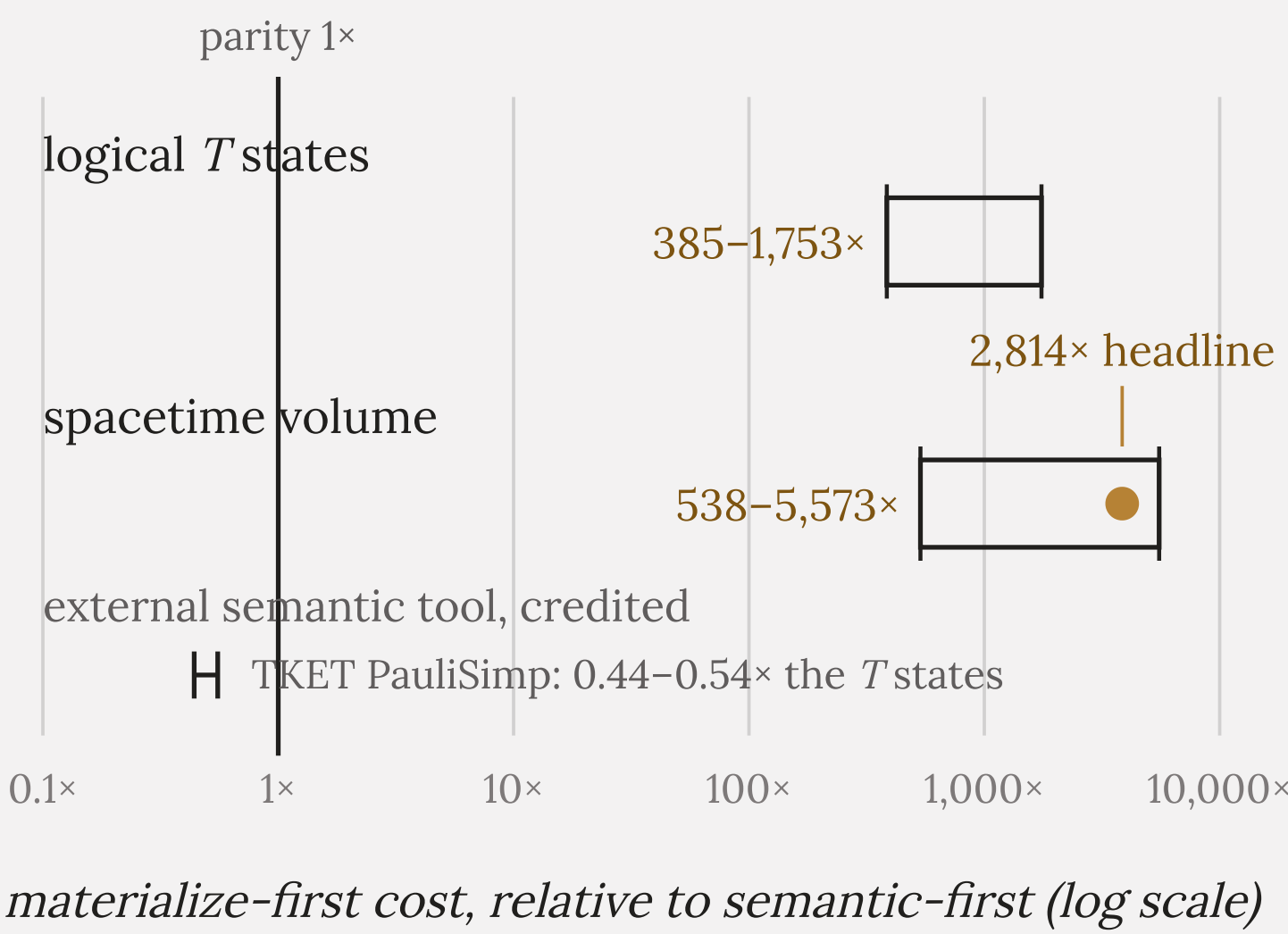
Once a committed record is *executed* rather than buffered, the committed channel pays a further $\log_2(\frac{m}{k})$ — the entropy of the compiler's own decision about which coordinates to commit early. A compiler that defers is handed that decision for free by the stream that follows.



Measured on an instrumented device under a preregistration frozen before the run. The toll is exactly zero when the commitment schedule is public.

05 What it costs downstream

Propagating one fixed total error budget through actual Clifford+ T synthesis and surface-code scheduling turns the representational gap into a fault-tolerant one. Over all 96 matched target / error / allocation / QEC / factory settings, no reversal occurs.



Outputs of checked angle synthesis and the disclosed scheduling models — not universal hardware predictions.

THE EXCHANGE RATE, IN MINIATURE

One QAOA cost layer, 3-regular MaxCut on 22 vertices — $m = 33$ edges at $\epsilon = 10^{-3}$. The layer is worth 252 bits, about **31 bytes**. A compiler that defers carries those 31 bytes across every cut. One that commits early buys the same information back at roughly 30 T gates per corrected coordinate: order **10³ magic states**.

06 Where the bound bites

Two hypotheses hold

Variational and Trotter cost layers: rounds contiguous in the stream, coefficients free to vary round to round.

One fails

Controlled-power ladders and phase estimation: contiguous, but round one already fixes the aggregate.

Clifford records — routing, swap, and the Pauli-frame insertions of randomized compiling itself — sit interior to the block and normalize away, so the sign-randomized and folded presentations obey the same floor.

1,080

matched cells, 9 representations
× 5 compilers

288

fixed-total-error fault-tolerant
estimates

1,728

external-baseline cells, bridge
loss reported apart

21,060

workload and negative-control
cells, six-factor ablation

Every headline number is generated from a normalized frozen registry of 24,748 records; one command reruns the data audits, tables, figures and tests, and an independent clean-room directory reconstructs the theorem, benchmark and resource-estimate subsets.

Full paper, artifacts and preregistrations: [link](#) or [contact here](#)

T A K E A W A Y

Randomize and lower late.

The information a compiler discards before aggregating cannot be recovered for free — it must be bought back in memory or in magic states, and buying it back early is strictly the more expensive of the two. Treat recoverability as a controlled variable in compilation and resource estimation.